

Experiment No. 3	
VOLTAGE AND CURRENT DIVISION PRINCIPLES	
1. Objective(s):	
The activity aims to determine the electrical characteristics of conductors using electric circuit configuration involving parallel and series resistances.	
2. Intended Learning Outcomes (ILOs):	
The students shall be able to: 2.1 Verify the relationship of voltage and current at varying resistance. 2.2 Determine the combine resistance of resistive loads when connected in series and parallel 2.3 Evaluate the voltages drop in electric conductors based on series resistors. 2.4 Apply the principles of voltage and current division in solving electrical circuits.	
3. Discussion:	
Resistance is the property of an electric circuit tending to prevent the flow of current and at the same time, causing electrical energy to convert into heat energy. Resistors are used where it is desired to introduce resistance into a circuit. Resistors may be connected in series and parallel. A series circuit is one in which the resistors or other electrical devices are connected end to end. The current in each part of the circuit is the same, but the total impressed voltage divides among the different elements of the circuit. This circuit is used as a voltage division in a series circuit hence, $V_1 = V_T R_1 / (R_1 + R_2) \text{ or } V_2 = V_T R_2 / (R_1 + R_2), \text{ ohms (Equation 1)}$ The total resistance of group resistors connected in series is equal to the sum of the individual resistances. $R_T = R_1 + R_2 + R_3 + \dots + R_N$ A parallel circuit is one of the terminals of each element is connected to a common point known as a junction and the other terminal of each element is connected to another junction. Each element has the same voltages across it, but the total current divides among the elements of the circuit. The circuit is used as current division in a parallel circuit hence, $I_1 = I_T R_2 / (R_1 + R_2) \text{ or } I_2 = I_T R_1 / (R_1 + R_2), \text{ ohms (Equation 2)}$ The equivalent resistance of the resistors connected in parallel is equal to the reciprocal of the sum of the reciprocal of the individual resistances. $1 / R_T = 1 / R_1 + 1 / R_2 + 1 / R_3 + \dots + 1 / R_N$ (Equation 3)	
4. Resources:	
PC-Based Circuit Trainer/Module with Prototyping Board and Connectors	1 set
Resistors , ¼ Watt (resistor values are to be selected by the students)	3 pcs
Digital VOM	2 units

5. Procedure:

Activity 1: Electric Conductors Represented in Series Configuration

- 1.1 Set the PC-Based trainer/module at desired voltage.
- 1.2 Connect the three (3) resistors at different resistances based on Figure 3.1.
- 1.3 Measure and record the cold resistance of each resistor and the equivalent resistance on Table 3.1 based on Figure 3.1.
- 1.4 Turn ON the supply voltage, then measure and record the source voltage and current in each resistor R_1 , R_2 , and R_3 on Table 3.2 using Figure 3.2. (This will give emphasize on the effect of temperature in an electric conductor due to current, otherwise known as hot resistance).
- 1.5 Measure and record the voltage drop of each resistor and the source voltage on Table 3.2 based on Figure 3.3.
- 1.6 Compute and record the voltages drop of each resistor on Table 3.3 using voltage division principle.
- 1.7 Turn OFF the source voltage, then calculate and record the percentage difference of voltage drop on Table 3.4.
- 1.8 Then perform activity no. 2.

CIRCUIT DIAGRAM:

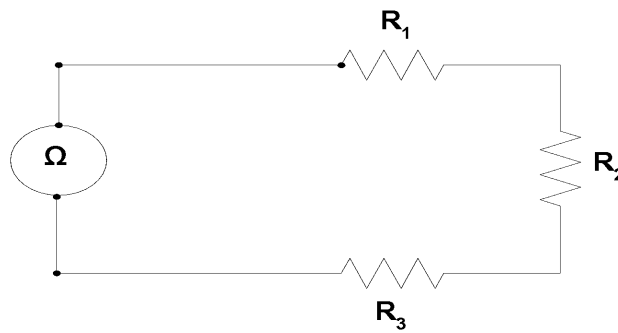


Figure 3.1. Series-Connected Resistors

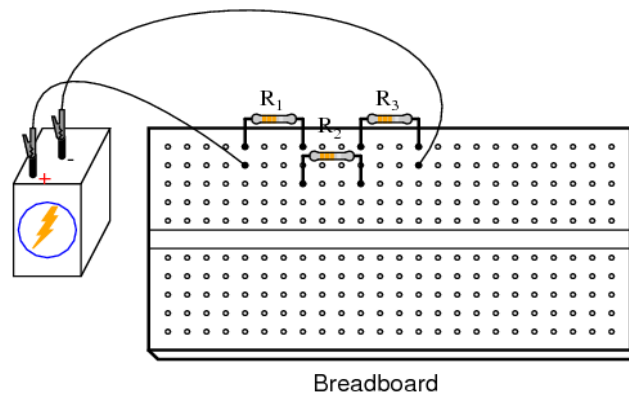


Figure 3.2. Electrical Diagram for Series Conductors using Breadboard

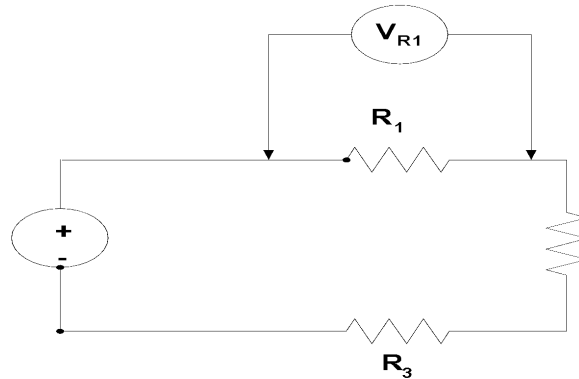


Figure 3.3. Voltage drop measurement of Series Conductors

Activity 2 - Electric Conductors Represented in Parallel Configuration

- 2.1 Connect the three (3) resistors based on Figure 3.4.
- 2.2 Then measure and record the cold resistance of each conductor and the equivalent resistance on Table 3.4 based on Figure 3.4.
- 2.3 Turn ON the source voltage, then measure and record the current drawn in each resistor on Table 3.4 based on Figure 3.
- 2.4 Turn OFF the supply voltage, and then compute the current drawn in each resistor using current division principle.

Activity 3 - Determining the Hot Resistance of Electric Conductors

- 2.5 Calculate the resistance of each resistor using voltage-current ratio and compare this to the measured resistance using Table 3.6 to obtain their percentage difference.

CIRCUIT DIAGRAM:

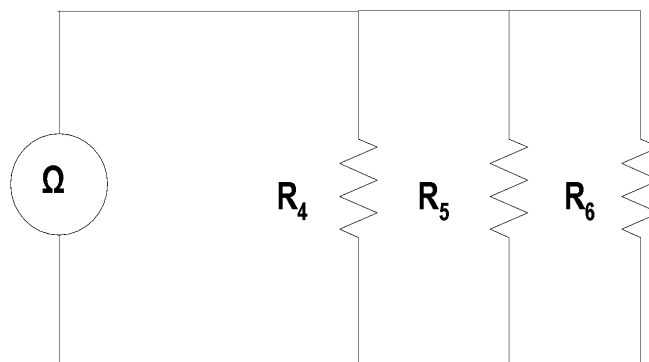


Figure 3.4 – Cold Resistance Measurement of Electric Conductors in

Parallel Connection

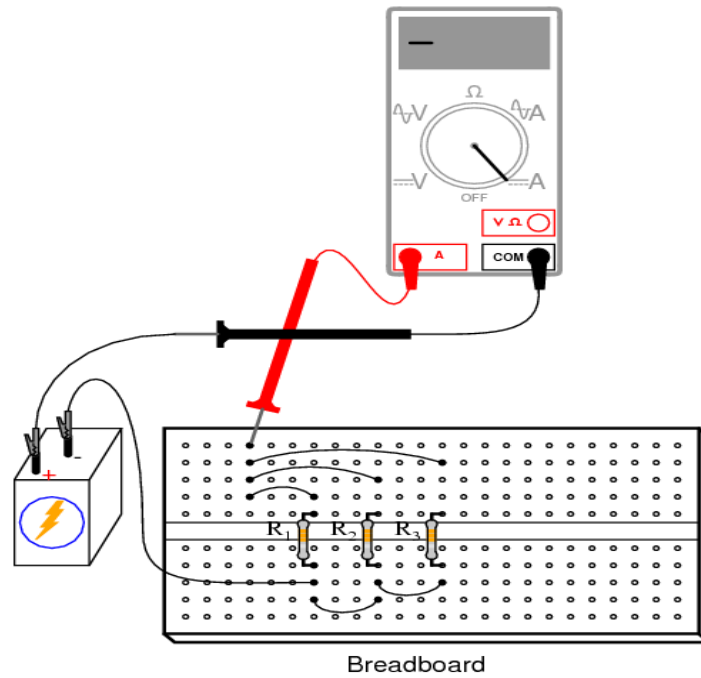


Figure: 3.5. Electrical Diagram for Parallel Conductors using Breadboard

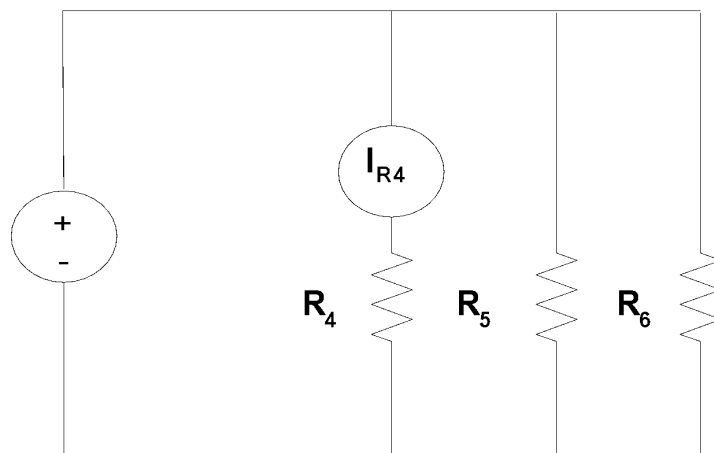


Figure:3.6. Current Measurement of Parallel Conductors

Course: EE 041	Experiment No.: 3																																								
Group No.: 2	Section: CPE21S2																																								
Group Members: Bungag, Mark Jhowell. Cruz, Elijah Condino, Niel Vincent Demain, Jaime Luis. Esguerra, Paula., Esteta, Jesse Matthew	Date Performed: Jul 30, 2026																																								
	Date Submitted:																																								
	Instructor: Christine Ann Alguero																																								
6. Data and Results:																																									
Activity 1 – Electric Conductors in Series Connection Table 3.1- Cold Resistance Data Summary <table border="1" style="margin-left: auto; margin-right: auto; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center;">R_1 (Ohms)</th> <th style="text-align: center;">R_2 (Ohms)</th> <th style="text-align: center;">R_3 (Ohms)</th> <th style="text-align: center;">R_{EQ} (Ohms)</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">100Ω</td> <td style="text-align: center;">98.02Ω</td> <td style="text-align: center;">102.8Ω</td> <td style="text-align: center;">300Ω</td> </tr> </tbody> </table> Table 3.2 – Measured Voltage Drop and Current Data Summary <table border="1" style="margin-left: auto; margin-right: auto; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center;">I_T (Amp)</th> <th style="text-align: center;">I_1 (Amp)</th> <th style="text-align: center;">I_2 (Amp)</th> <th style="text-align: center;">I_3 (Amp)</th> <th style="text-align: center;">V_T (Volts)</th> <th style="text-align: center;">V_1 (Volts)</th> <th style="text-align: center;">V_2 (Volts)</th> <th style="text-align: center;">V_3 (Volts)</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">0.05A</td> <td style="text-align: center;">0.05A</td> <td style="text-align: center;">0.05A</td> <td style="text-align: center;">0.05A</td> <td style="text-align: center;">15V</td> <td style="text-align: center;">5V</td> <td style="text-align: center;">5V</td> <td style="text-align: center;">5V</td> </tr> </tbody> </table> Table 3.3 – Voltage Drop Data Summary using VDP <table border="1" style="margin-left: auto; margin-right: auto; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center;">$V_t R_1/(R_1+R_2+R_3)$ Volts</th> <th style="text-align: center;">$V_t R_2/(R_1+R_2+R_3)$ Volts</th> <th style="text-align: center;">$V_t R_3/(R_1+R_2+R_3)$ Volts</th> <th style="text-align: center;">ΣV, Volts</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">$12(100)/(300) = 5V$</td> <td style="text-align: center;">$12(100)/(300) = 5V$</td> <td style="text-align: center;">$12(100)/(300) = 5V$</td> <td style="text-align: center;">15V</td> </tr> </tbody> </table> Table 3.4. Percent Difference in Voltages Drop <table border="1" style="margin-left: auto; margin-right: auto; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center;">V_T (Ohms)</th> <th style="text-align: center;">V_1 (Ohms)</th> <th style="text-align: center;">V_2 (Ohms)</th> <th style="text-align: center;">V_3 (Ohms)</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">100%</td> <td style="text-align: center;">33.33%</td> <td style="text-align: center;">33.33%</td> <td style="text-align: center;">33.33%</td> </tr> </tbody> </table>		R_1 (Ohms)	R_2 (Ohms)	R_3 (Ohms)	R_{EQ} (Ohms)	100Ω	98.02Ω	102.8Ω	300Ω	I_T (Amp)	I_1 (Amp)	I_2 (Amp)	I_3 (Amp)	V_T (Volts)	V_1 (Volts)	V_2 (Volts)	V_3 (Volts)	0.05A	0.05A	0.05A	0.05A	15V	5V	5V	5V	$V_t R_1/(R_1+R_2+R_3)$ Volts	$V_t R_2/(R_1+R_2+R_3)$ Volts	$V_t R_3/(R_1+R_2+R_3)$ Volts	ΣV , Volts	$12(100)/(300) = 5V$	$12(100)/(300) = 5V$	$12(100)/(300) = 5V$	15V	V_T (Ohms)	V_1 (Ohms)	V_2 (Ohms)	V_3 (Ohms)	100%	33.33%	33.33%	33.33%
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Activity 2 – Electric Conductors in Parallel Connection

Table 3-5. Cold Resistance Data Summary

R_1 (Ohms)	R_2 (Ohms)	R_3 (Ohms)	R_{EQ} (Ohms)
100Ω	98.02Ω	102.8Ω	33.41Ω

Table 3-6. Measured Voltage Drop and Current Data Summary

I_T (Amp)	I_1 (Amp)	I_2 (Amp)	I_3 (Amp)	V_T (Volts)	V_1 (Volts)	V_2 (Volts)	V_3 (Volts)
0.4489A	0.15A	0.153A	0.1459A	15V	15V	15V	15V

Table 3-7. Current Data Summary using CDP

I_1	I_2	I_3	ΣI , Amps
15V/100 ohms = 0.1500 A	15/98.02 = 0.153 A	15/102.8= 0.145 A	0.4490A

Table 3.8. Percent Difference in Electric Current

	Calculated Current, Amp	Measured Current, Amp	%Difference
Current, I_1	0.1500 A	0.151 A	0.66
Current, I_2	0.153 A	0.150 A	1.98
Current, I_3	0.145 A	0.147 A	1.37
ΣI , Amps	0.4489A	0.4489A	0

Activity 3 - Determining the Hot Resistance of Electric Conductors

Table 3.7 - Calculating the Hot Resistance of Electric Conductor using Voltage-Current Ratio

V_T / I_T	V_T / I_{R4}	V_T / I_{R5}	V_T / I_{R6}	$\Sigma R, \text{ ohms}$
99.4 Ω	100 Ω	102.04 Ω	33.41 Ω	301.38 Ω

Table 3.8 –Percent Difference in the Resistance of Conductor

R_T	R_4	R_5	R_6
0%	-0.66%	1.98%	-0.74%

$$\text{Percentage difference in resistance} = \frac{\text{Hot Resistance} - \text{Cold Resistance} (100\%)}{\text{Hot Resistance}}$$

7. Question and Answer

- State the advantage of using current and voltage divider principles.
 - The main advantage of using current and voltage divider principles is that they make circuit analysis much simpler by calculating the current or voltage in any component without solving the needed parameters for the entire circuit. It saves time and simplifies the calculation of troubleshooting electrical and electronic circuits.
- Strings of Christmas lights are sometimes made of miniature lamps connected in series. For a twenty-four 240-volt lamp circuit, what is the voltage across each lamp? If one lamp is removed, what would happen? Explain why?
 - The voltage of each lamp is 10V. Since the total voltage is 240V and there are 24 lamps in total, we divide it and get the answer of 10V. If one lamp is removed, the entire circuit breaks. This is because in a series circuit, all lamps are connected and share the same current path. If one is removed, no current will flow through the remaining lamps.
- A piece of copper wire is cut into N equal parts. These parts are connected in parallel. What is the ratio of resistance of the parallel combination compared to the resistance with the original length?
 - When they are cut into N equal parts, the length of each segment becomes 1/N of the original wire, making each piece's resistance equal to R/N. Connecting these N equal segments in parallel divides the resistance of a single segment by N once again. This gives the equivalent parallel resistance of $R(\text{parallel}) = R/N^2$. The resistance is 1: N².
- What experimental procedure is needed to determine the hot and cold resistance of electric conductors?
 - To determine cold resistance, measure the unpowered conductor at room temperature using an ohmmeter. To determine hot resistance, connect the conductor to its rated operating voltage, allow it to reach full operating temperature, and simultaneously measure the active voltage and current
- Why are electric conductors designed at different ampere ratings (ampacities)? Support your answer with Table of "Conductor Size and Current Ratings"

- Electric conductors are designed with different ampere ratings so that they safely handle different levels of current without overheating. Because electrical resistance causes current to generate heat, thick wires with large cross-sectional areas provide lower resistance to handle larger currents safely.

Conductor Size (AWG)	Maximum Ampacity (60°C Insulation) (A)	Maximum Ampacity (75°C Insulation) (A)
14	15	20
12	20	25
10	30	35
8	40	50
6	55	65

6. Two rheostats of resistance 60 ohms and 120 ohms are connected in parallel and joined to the battery of negligible internal resistance. If the current in the 60-ohm rheostat is 0.2 ampere, calculate the current in the 120-ohm rheostat and the battery voltage using the current division technique.

(Show your solution in a separate answer sheet).

$R_1 = 60\Omega$ $R_2 = 120\Omega$ $R_T = 40\Omega$
 $I_1 = 0.2\text{ A}$ $I_2 = ?$ $I_2 = \frac{60}{60+120}(I_T)$ $I_1 = \frac{120}{60+120}(I_T)$
 $V_1 = V_2$ $I_2 = \frac{60}{180} I_T$ $0.2 = \frac{2}{3}(I_T)$
 $I_2 = I_T/3$ $(0.2)(3/2) = I_T$
 $I_T = 0.3$
 $I_2 = 0.3/3$
 $I_2 = 0.1\text{ A}$
 $V_T = I_T/R_T$
 $V_T = 0.3/40$
 $V_T = 0.0075\text{ V}$
 ∴ The current in the 120Ω rheostat is 0.1 A and the voltage of the battery is 0.0075 V

8. Conclusion:				
In this lab, we determined the electrical characteristics of conductors by analyzing series and parallel circuit configurations. The collected data verified Ohm's law, proving that current is directly proportional to voltage and inversely proportional to resistance. When testing combined resistances, we confirmed that a series connection increases the total resistance because the resistors are added directly together. In contrast, a parallel connection decreases total resistance, following the reciprocal formula. We also successfully applied the principles of voltage and current division. In the series circuits, we observed that the total voltage drops across each resistor based on its resistance value while the current stays constant throughout. For the parallel circuits, the voltage remained the same across all branches, while the total current divided inversely through each path. Ultimately, these experimental results validate the mathematical principles used to analyze and solve resistive networks.				
9. Assessment:				
CRITERIA	BEGINNER 1	ACCEPTABLE 2	PROFICIENT 3	SCORE
I. Laboratory Skills				
Manipulative Skills	Members do not demonstrate needed skills.	Members occasionally demonstrate needed skills	Members always demonstrate needed skills.	
Experimental Set-up	Members are unable to set-up the materials.	Members are able to set-up the materials with supervision.	Members are able to set-up the material with minimum supervision.	
Process Skills	Members do not demonstrate targeted process skills.	Members occasionally demonstrate targeted process skills.	Members always demonstrate targeted process skills.	
Safety Precautions	Members do not follow safety precautions.	Members follow safety precautions most of the time.	Members follow safety precautions at all times.	
II. Work Habits				
Time Management / Conduct of Experiment	Members do not finish on time with incomplete data.	Members finish on time with incomplete data.	Members finish ahead of time with complete data and time to revise data.	
Cooperative and Teamwork	Members do not know their tasks and have no defined responsibilities. Group conflicts have to be settled by the teacher.	Members have defined responsibilities most of the time. Group conflicts are cooperatively managed most of the time.	Members are on tasks and have defined responsibilities at all times. Group conflicts are cooperatively managed at all times.	

Neatness and Orderliness	Messy workplace during and after the experiment.	Clean and orderly workplace with occasional mess during and after the experiment.	Clean and orderly workplace at all times during and after the experiment.	
Ability to do independent work	Members require supervision by the teacher.	Members require occasional supervision by the teacher.	Members do not need to be supervised by the teacher.	
Other Comments/Observations:			Total Score	
			$Rating = \frac{(Total\ Score)}{24} \times 100$	
10. Reference(s): Hayt.W. (2012). <i>Engineering circuit analysis (8th ed.)</i> .New York, McGraw Hill				